

A wind tunnel study of the effects of adjacent buildings on near-field pollutant dispersion from rooftop emissions in an urban environment

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ABSTRACT

This paper presents results from a wind tunnel study of near-field pollutant dispersion from rooftop emissions of two multiple building configurations. The configurations mainly consisted of an emitting building in the presence of an upstream and a downstream building. The various parameters that were varied include: stack height (h_s), stack location (X_s), spacing between upstream and emitting building (S_1), spacing between downstream and emitting building (S_2) and exhaust momentum ratio (M). Gas concentrations were measured at various building surfaces using a gas chromatograph. The wind tunnel dilutions were also compared to ASHRAE, 2007 and 2011 models. Results show that a taller upstream and a taller downstream building inhibit the plume from dispersing, thereby increasing the pollutant concentrations on the roof of the emitting building and leeward wall of the upstream building. In general, the spacing between the upstream and emitting buildings, besides the heights of each building were found to be critical parameters influencing the plume characteristics. ASHRAE, 2007 predictions were found to be overly conservative for the isolated building, while ASHRAE, 2011 estimates compared well with experimental data for a few cases. Safe placement of stack and intake on various building surfaces to avoid plume re-ingestion are suggested based on this study.

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1. Introduction

Pollutants released from a rooftop stack can re-enter the building from which they are released or even enter a neighbouring building (Stathopoulos et al., 2008). In an urban environment, buildings are closely spaced as shown in Fig. 1, which depicts a view of downtown Toronto, Canada as seen from the CN tower. Unfortunately, the state-of-the-art is not fully developed to accurately assess the flow and concentration of pollutants through such a densely populated urban layout. Mavroidis and Griffiths, 2001 performed a flow visualization study (Fig. 2) for smoke dispersing through an array of obstacles, representing buildings. Their study showed that the plume geometry was affected as the spacing between the obstacles changed. However, no detailed study has been made to understand the pollutant flow in an urban environment. Most studies have focused on isolated building configurations that seldom exist in the built environment (e.g. Halitsky, 1963; Wilson, 1979 etc.). Near-field plume dispersion is greatly influenced by adjacent buildings as opposed to far-field problems where atmospheric turbulence is greater (Saathoff et al.,

2009). There are many studies that have focussed on pollutant dispersion in street-canyons using wind tunnel and CFD simulations (e.g. Wedding et al., 1977; Chang and Meroney, 2000, 2001, 2003; Meroney, 2010), with few studies on the application of ASHRAE models on micro-scale pollutant dispersion problems (Stathopoulos et al., 2004, 2008). Recently, Hajra et al., 2011 carried out a detailed investigation of the effects of upstream buildings on near-field pollutant dispersion. The effect of downstream buildings of different geometries on effluent dispersion from rooftop emissions was performed by Hajra and Stathopoulos, 2012 more recently. The results from both these studies provided design guidelines for the safe placement of stack and intake on various building surfaces. The next step would be to include the effects of urban environment in terms of additional buildings placed in the vicinity of the emitting building which would affect the wind and pollutant flow. In order to accomplish this, the present study aims to extend the ongoing investigation to multiple building configurations consisting of a building placed upstream and another building placed downstream of an emitting building.

Efforts were made by Li and Meroney, 1983 to distinguish between near-field and far-field dispersion problems. They defined the "near-wake" region as $x/H < 5$, where x is the distance of the receptor from the source and H is the height of the building. Similarly, Wilson et al., 1998 defined near-field to be the distance

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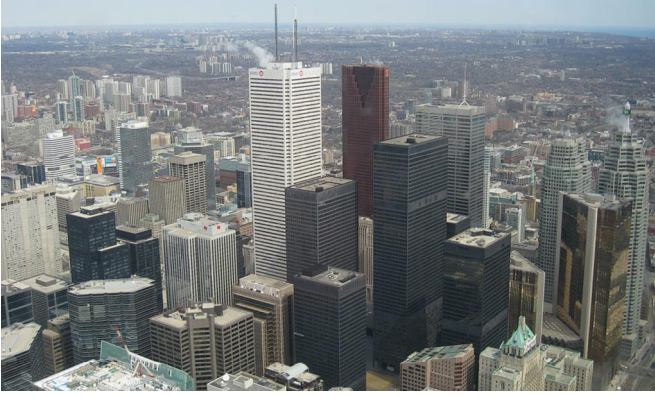


Fig. 1. View of downtown Toronto, Canada; picture taken from CN Tower.

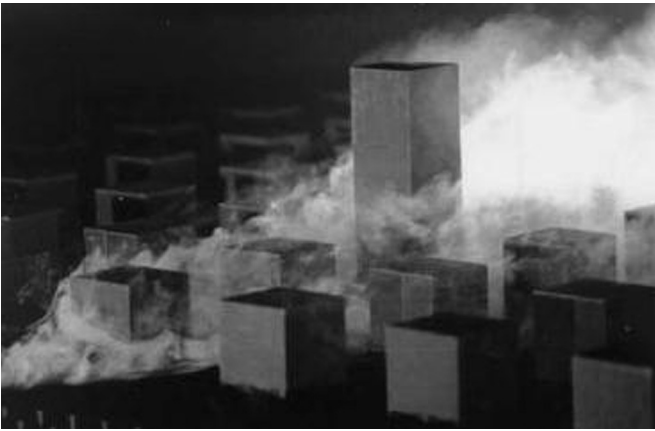


Fig. 2. Smoke dispersing through an array with an in-line configuration and a spacing of $S/H=1.5$, with a taller obstacle ($H=3W$) located in the 3rd row of the array (from Mavroidis and Griffiths, 2001).

within the “recirculation region” from the source which is estimated from the dimensions of the building perpendicular to wind direction. The results of Wilson’s study are still being used in the semi-Gaussian ASHRAE, 2007, 2011 models.

Other available dispersion models such as ADMS, SCREEN and AERMOD were not used for this study since they are incapable of simulating the turbulence caused by nearby buildings, and hence cannot accurately predict pollutant concentrations on building roofs (Stathopoulos et al., 2008). In fact, Riddle et al., 2004 suggested that “such atmospheric dispersion packages are not able to assess the local effects of a complex of buildings on the flow field and turbulence, and whether gas will be drawn down amongst the buildings”. However, ASHRAE, 2007, 2011 have been used for the present study since they are capable of assessing dilutions on rooftop receptors, based on the recirculation zone formed in the building wake.

Section 2 of this paper describes the air and pollutant flow for different building configurations followed by a description of ASHRAE, 2007, 2011 models in Section 3. The experimental procedure and the various building configurations examined have been discussed in Sections 4 and 5 respectively. Results and discussion have been presented in Section 6. This is followed by design guidelines for safe placement of stack and intake on various building surfaces, as well as a summary of findings in Section 7. The conclusions of this study have been presented in Section 8, besides an appendix illustrating the application of ASHRAE, 2007, 2011 models.

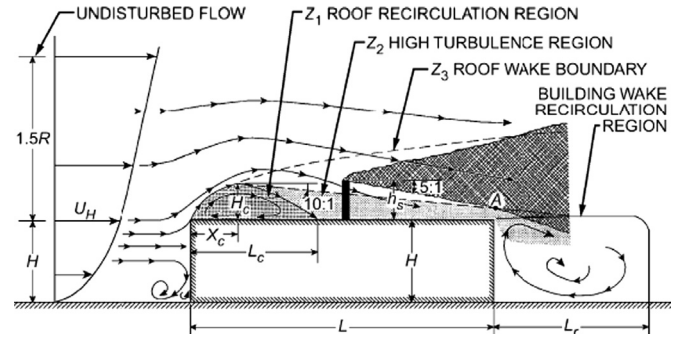


Fig. 3. Design procedure for required stack height to avoid contamination (from Wilson, 1979).

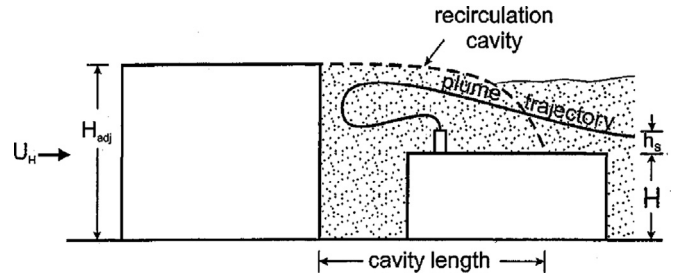


Fig. 4. Recirculation cavity for a taller upstream building (from Wilson et al., 1998).

2. Air and pollutant flow around buildings

Based on a series of experiments, Wilson, 1979 showed that the size of the recirculation region (shown as L_r in Fig. 3) formed in the wake of a building is estimated by using the building dimensions perpendicular to wind direction:

$$L_r = B_s^{0.67} B_L^{0.33} \quad (1)$$

where:

L_r is the zone of recirculating flow formed in the building wake (m),

B_s is the smaller building dimension perpendicular to wind direction (m),

B_L is the larger building dimension perpendicular to wind direction (m).

Wilson showed that turbulence due to the building occurs up to about 1.5 times ‘R’ from the roof of the building, where ‘R’ is the scaling length for roof flow patterns. The value of ‘R’ is obtained from Eq. (1), by replacing ‘ L_r ’ by ‘R’. He suggested that the pollutants released from a rooftop stack form a triangle (in two dimensions) with the edges at 5:1 away from the plume centre-line. Additionally, a recirculation length (L_c) also forms on the roof besides L_r in the wake for a longer building, as shown in Fig. 3. However, Wilson et al., 1998 was able to show that the plume trajectory changes in the presence of an upstream building, as shown in Fig. 4. They showed that the wake recirculation cavity of the upstream building brought the plume towards the leeward wall of the upstream building and the roof of the emitting building thereby increasing effluent concentrations on the emitting building. Similar observations were made by Stathopoulos et al., 2004 during field measurements at Concordia University. According to Wilson et al., 1998, the presence of a taller downstream building prevented the plume from dispersing along the roof of the emitting building with a small portion of the plume also escaping from the sides as “side-leakage” and over the roof of the

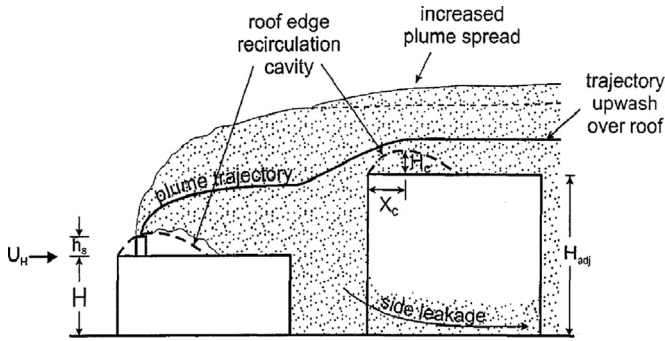


Fig. 5. Side leakage phenomenon for taller downstream building (from Wilson et al., 1998).

downstream building as upwash, as shown in Fig. 5. However, most studies were limited to only a few building configurations, and no detailed studies by changing different parameters was carried out. The air and pollutant flow in the presence of upstream buildings and in the presence of downstream buildings is much better understood following detailed studies carried out by Hajra et al., 2011 and Hajra and Stathopoulos, 2012. The subsequent section describes the ASHRAE models which have been used in the present study.

3. ASHRAE models

This section describes the semi-Gaussian ASHRAE, 2007, 2011 models. Both models have two methods namely: Geometric design method and the Gaussian plume equations. The geometric design method is a qualitative approach and is mainly used to assess the minimum stack height to avoid plume re-ingestion through the leeward wall of the emitting building. The Gaussian plume equation is a quantitative technique used to estimate rooftop dilutions. The geometric design method has remained unchanged in ASHRAE, 2007, 2011 models, while changes have been suggested in the Gaussian approach, as discussed further herein.

3.1. Geometric design method

The geometric design method assumes that the plume released from a stack follows a triangular path with the sides at 5:1 away from the centreline (Fig. 3).

The dimensions of flow re-circulation zones that form on the building are expressed in terms of L_r :

$$H_c = 0.22L_r \quad (2)$$

$$X_c = 0.5L_r \quad (3)$$

$$L_c = 0.9L_r \quad (4)$$

where:

- H_c is the maximum height of the roof recirculation zone (m),
- X_c is the distance from the leading edge to H_c (m),
- L_c is the length of the roof recirculation zone (m).

The boundary of the high turbulence region is defined by a line with a slope of 10:1 extending from the top of the leading edge separation bubble. Therefore, the geometric design method can only be used to estimate the minimum stack height that can avoid the recirculation length (L_r) formed in the wake of the building. However, for assessing plume dilutions at a rooftop receptor, Gaussian plume equations are used.

3.2. Gaussian plume equations

ASHRAE, 2007, 2011 have made several changes in estimating plume dilutions. Each model is discussed separately.

3.2.1. ASHRAE 2007

The plume dilutions are estimated by calculating certain parameters that include the effective height of the plume (h) above the roof:

$$h = h_s + h_r - h_d \quad (5)$$

where:

- h_s is stack height (m),
- h_r is plume rise (m),
- h_d is the reduction in plume height due to entrainment into the stack wake during periods of strong winds (m).

Plume rise is calculated using the formula of Briggs, 1984:

$$h_r = 3\beta d_e (V_e/U_H) \quad (6)$$

where:

- d_e is the stack diameter (m),
- V_e is the exhaust velocity (m/s),
- U_H is the wind speed at building height (m/s) and β is the stack capping factor. The value of β is 1 for uncapped stacks and 0 for capped stacks.

To account for the stack downwash caused Wilson et al., 1998 recommended a stack wake downwash adjustment h_d , defined as:

$$h_d = d_e(3 - \beta V_e/U_H) \quad (7)$$

For $V_e/U_H > 3.0$ there is no stack downwash ($h_d = 0$).

Dilution at roof level in a Gaussian plume emitted at the final rise plume height of h is:

$$D_r = 4(U_H/V_e)(\sigma_y/d_e)(\sigma_z/d_e)\exp(\zeta^2/2\sigma_z^2) \quad (8)$$

where:

$$\zeta = h - H_c = 0 \text{ if } h < H_c$$

ζ is the vertical separation between 'h' and H_c .

It may be mentioned that D_r is also expressed as a ratio of exhaust concentration (C_e) to receptor concentration (C_r). According to Hajra and Stathopoulos, 2012 " C_r is proportional to the pollutant emission rate Q and not exhaust concentration (C_e) since the latter may be altered by addition of air without affecting receptor concentrations".

The plume equations are as follows:

$$\sigma_y/d_e = 0.071(t_{avg}/2)^{0.2}(X/d_e) + \sigma_o/d_e \quad (9)$$

$$\sigma_z/d_e = 0.071(X/d_e) + \sigma_o/d_e \quad (10)$$

Dilutions calculated from Eq. (8) have been converted to normalised dilutions using the formulations of Wilson, 1979 for comparison with previous studies.

$$D_{\text{normalised}} = (D_r Q)/(U_H H^2) \quad (11)$$

where:

- $Q = \pi d_e^2 V_e / 4$ is the volumetric flow-rate (m^3/s)
- H is the height of the building (m).

3.2.2. ASHRAE 2011

ASHRAE, 2011 has recently been introduced due to discrepancies obtained for ASHRAE, 2007 and experimental data from

previous studies for isolated building cases (Stathopoulos et al., 2008; Hajra et al., 2010). New formulations for estimating plume rise (h_r), plume spread parameters (σ_y and σ_z) and dilution for shorter time periods have been suggested. Plume rise (h_r) is estimated as:

$$h_r = \min(\beta h_x \beta h_f) \quad (12)$$

where: h_x and h_f are estimated as

$$h_x = \left(\frac{3V_e^2 d_e^2 X}{4\beta_j^2 U_H^2} \right)^{1/3} \quad (13)$$

$$h_f = \frac{0.9[(V_e^2 d_e^2 / 4)(U_H / U_*)]^{0.5}}{\beta_j U_H} \quad (14)$$

where:

U_* is the friction velocity (m/s),

β_j is termed as the jet entrainment coefficient and is calculated as

$$\beta_j = \frac{1}{3} + \frac{U_H}{V_e} \quad (15)$$

The logarithmic wind profile equation is

$$U_H / U_* = 2.5 \ln(H / Z_o) \quad (16)$$

where:

Z_o is the surface roughness length (m)

The plume rise as per ASHRAE, 2007 (Eq. (6)) is a function of the exhaust momentum ratio (M) and stack diameter (d_e) while the 2011 version takes account of the effects of wind velocity profile and stack-receptor distance (X). The formulations suggested by Cimoreli et al., 2005 have been used to estimate the plume spread parameters.

$$\sigma_y = (i_y^2 X^2 + \sigma_0^2)^{0.5} \quad (17)$$

$$\sigma_z = (i_z^2 X^2 + \sigma_0^2)^{0.5} \quad (18)$$

$$i_y = 0.75 i_x \quad (19)$$

$$i_z = 0.5 i_x \quad (20)$$

$$i_x = [0.24 + 0.096 \log_{10}(Z_o) + 0.016(\log_{10} Z_o)^2] [\ln(30/Z_o) / \ln(Z/Z_o)] \quad (21)$$

where:

i_x , i_y and i_z are the turbulence intensities in x, y and z directions, σ_0 is the initial source size and is set equal to $0.35 d_e$ (m), Z is the height of the building (m).

The source size (σ_0) is defined as a function of M and d_e in ASHRAE, 2007 while ASHRAE, 2011 defines σ_0 as a function of d_e . ASHRAE, 2011 states (in an example) that the lowest dilution value must be taken, based on calculations performed for Z_o , $0.5Z_o$ and $1.5Z_o$. ASHRAE, 2007 states "For the case of both stack tip and air intake in the same wind recirculation zone, assume the D_r values for 2 min averages also apply for all averaging times from 2 to 60 min." As per ASHRAE, 2011, the dilution calculated from Eq. (8) is equivalent to 10–15 min averaging time, and hence for uniformity, calculations as per ASHRAE, 2007 (Eq. (9)) have considered $t_{avg} = 15$ min in the present study. However, for shorter averaging times, ASHRAE, 2011 suggests the following formula:

$$(D_r)' = D_r (t_{avg} / 15)^{0.2} \quad (22)$$

Table 1

Experimental parameters used in the present study.

Experimental parameters	Present study (wind tunnel values)
Model scale	1:200
Boundary layer depth (δ)	95 cm
Wind speed at building height (U_H)	6.2 m/s
Power law exponent (α)	0.31
Upstream terrain	Urban
Velocity at gradient height (V_g)	14.2 m/s
Roughness length of upstream exposure	3.5 mm
Longitudinal integral scale	0.4 m
Stack diameter (d_e)	0.3 cm
Averaging time (t_{avg})	1 min
Upstream turbulence at building height (%)	23 ^a , 17 ^b

^a Low rise building of 15 m.

^b Intermediate building of 30 m.

where:

$(D_r)'$ is the dilution estimated for a shorter averaging time t_{avg} ,

t_{avg} is the averaging time in minutes,

D_r is the dilution calculated as per Eq. (8).

Indeed, the introduction of averaging time in ASHRAE, 2011 is an important step towards the improvement of ASHRAE model.

4. Wind tunnel experimentation and simulations

The present study examines wind tunnel data for four different configurations (two isolated cases and two multiple building configurations), three different stack heights (h_s) of 1, 3 and 5 m and exhaust momentum ratios (M) of 1, 2 and 3 at wind angle of 0° . The configurations consist of a building placed upstream and a building placed downstream of an emitting building. A low and intermediate emitting building have been used. The building models have a flat roof, with receptors located on the roof, leeward and windward walls. Design guidelines on the safe placement of stack and intake to avoid plume re-entrainment, and suggestions for improving ASHRAE models have been made, based on this study.

The experiments were performed at the Boundary Layer Wind Tunnel Laboratory at Concordia University, which is 1.8 m square in section and 12.2 m in length (Stathopoulos, 1984). Spires that act as vortex generators, and coarse roughness elements (5 cm cubes) staggered 6 cm from each other, were used to generate a thick atmospheric boundary layer. A power law exponent (α) of 0.31, which according to ASHRAE, 2009 corresponds to an urban terrain, was used for the study. The velocity and turbulence profiles were measured using a Cobra Probe, whose accuracy is ± 0.5 m/s up to turbulence intensity values of about 30% (Turbulent Flow Instrumentation, 2008). A scale of 1: 200 was used for the study. Additional boundary layer measurements are mentioned in Table 1. Experimental details can also be found in Stathopoulos et al., 2008 and Hajra and Stathopoulos, 2012.

The roof of the tunnel was adjusted to ensure that the longitudinal static pressure gradient was negligible. A mixture of SF₆ and Nitrogen was released from a rooftop stack of 3 mm diameter and stack height (h_s) of 1, 3 and 5 m. The exhaust momentum ratio (M), which is defined as the ratio of exhaust velocity (V_e) to wind velocity at building height (U_H), was varied from 1, 2 and 3. In general, exhaust momentum is a product of the density and velocity of the gas. However, for non-buoyant gases the densities of gas and air are nearly equal, and hence M reduces to a ratio of velocities. Concentration of tracer gas was carried out once the wind tunnel was stable after about 5 min. A syringe sampler,

which could collect the tracer gas samples in 1 min, was connected to various receptors via tubing's underneath the test section. ASHRAE, 2007 assumes an averaging time of 2 min in the wind tunnel equivalent to an hourly field averaging time. Generally, a receptor located in a high turbulence region may require higher sampling time as opposed to a receptor in a low turbulence zone. Since, in the present study the syringe sampler is capable of collecting gas samples for upto 1 min, an averaging time of 1 min was used. This difference in collection time did not have any impact on the accuracy of the results, as discussed in detail by Saathoff et al., 1995 and Stathopoulos et al., 2004. Generally, it is necessary to collect the samples for long durations until a steady-state average concentration is obtained. According to Snyder, 1981, the error (ϵ) involved in experimental measurements of pollutant dispersion in the wind tunnel is related to the averaging time (t_{avg}), boundary layer depth (δ) and the free stream velocity (U_∞) as:

$$t_{avg} = 4\delta / (U_\infty \epsilon^2) \quad (23)$$

For $t_{avg} = 1$ min, $\delta = 95$ cm, and $U_\infty = 14.2$ m/s, an error (ϵ) of 6.6% was obtained, which is generally considered to be low for near-field pollutant dispersion studies. The efficient ventilation facility of the laboratory ensured that there was no accumulation of SF₆ in the laboratory that could affect the measurements, as shown from previous studies at Concordia University (Saathoff et al., 2009). A VARIAN 3400 Gas Chromatograph whose precision is 5% and measurement resolution equal to one, was used to estimate the concentration of the gas samples (Stathopoulos et al., 2004).

According to Snyder, 1981, the following criteria need to be satisfied for modelling non-buoyant plume exhaust:

- Geometric similarity
- Building Reynolds Number > 11,000
- Stack Reynolds Number > 2000
- Similarity of wind tunnel flow with atmospheric surface layer
- Equivalent stack momentum ratio.

The building and stack Reynolds number were measured to be 20,000 and 1800 respectively. Saathoff et al., 1995 suggested that it is generally not possible to achieve the required stack Reynolds number because a trip cannot be placed around small diameter stacks. Previous studies carried out by Stathopoulos et al., 2008 have shown that even if the stack Reynolds number is somewhat less than 2000, it does not affect the accuracy of the measurements. Further discussion on stack Reynolds number and averaging time criteria are also available in Hajra et al., 2010.

5. Configurations investigated

Four different configurations were tested to assess near-field plume characteristics of adjacent buildings. The dimensions of each building model, as well as the recirculation lengths estimated from ASHRAE and ADMS approaches are presented in Table 2. ADMS is a dispersion model used primarily in UK, and uses the formulations of Fackrell and Pearce, 1981.

$$L_r = \frac{1.8W}{[(L/H)^{0.3}(1 + 0.24W/H)]} \quad (0.3 \leq L/H \leq 3.0) \quad (24)$$

Where:

W is the width/across wind dimension of the building (m),
 L is the length/along wind dimension of the building (m),
 H is the height of the building (m).

However, studies by Hajra et al., 2011 have shown that dilution predictions by Eq. 24 produce overly conservative results. Hence,

Table 2

Full scale dimensions of building models tested.

Building	Height (m)	Width (m)	Breadth (m)	Recirculation length	
				ASHRAE (Eq. (1))	ADMS (Eq. (24))
B ₁	15	50	50	22.3	35.9
B ₂	30	50	50	35.5	55.1
B ₃	30	50	30	35.5	50.0
B ₄	54	50	15	51.2	105.6

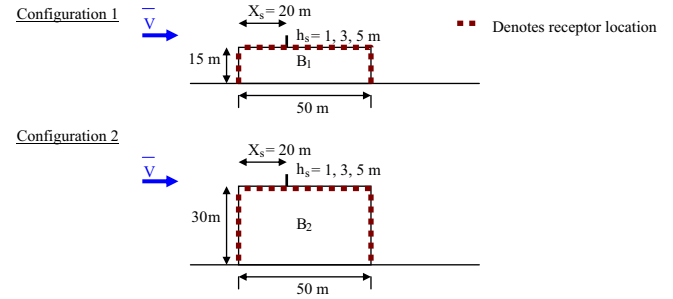


Fig. 6. Configurations 1 and 2: Sources located on low and intermediate height buildings.

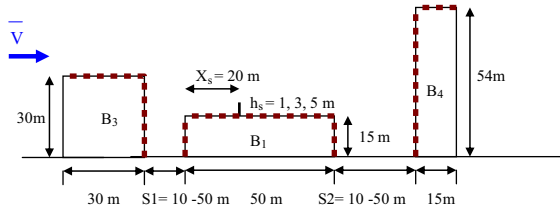
ADMS was not used for the present study. Buildings B₁ (low building) and B₂ (intermediate building) were used as emitting buildings (buildings with rooftop stack) for the present study. Figs. 6 and 7 show the receptor locations and dimensions of each building configuration. The receptors were located 5 m apart along the building centre line and not laterally over the various surfaces. On the emitting buildings (B₁ and B₂) the receptors were located on all surfaces (roof; leeward and windward walls). However, the location of receptors was restricted to the leeward wall and roof of B₃; windward wall and roof of B₄, since these surfaces were more affected as per preliminary flow visualisation investigations.

The width (across wind dimension) of all buildings was equal (50 m) since previous studies have shown that narrow adjacent (upstream/downstream) buildings do not increase plume concentrations on the building surfaces (Hajra et al., 2011). Additionally, wind angle of only 0° was considered because previous studies have shown that this is a more critical case for emissions from rooftop stacks than wind azimuths of 22.5° and 45° (Chui and Wilson, 1988; Hajra and Stathopoulos, 2012). The focus of this study was to assess dilutions within the building recirculation length. Therefore, the spacing (S1 and S2) between the buildings was varied between 10 m and 50 m, since the lowest and highest values of the recirculation length of the building is 22.3 m and 51.2 m respectively, as shown in Table 2. It may be mentioned that the spacing between the buildings was varied uniformly for any given test (S1=S2). The stack location from the upwind edge of the emitting building (X_s) was varied from 0 to 20 m.

6. Results and discussion

The results and discussion presented in this section are divided into three subsections. The first two subsections are devoted to dilutions on various building surfaces for Configurations 3 and 4 for a spacing (S1=S2) of 20 m (Figs. 8–12). The third subsection describes the effect of spacing between buildings on dilutions at various building surfaces (Figs. 13–15).

Configuration 3



Configuration 4

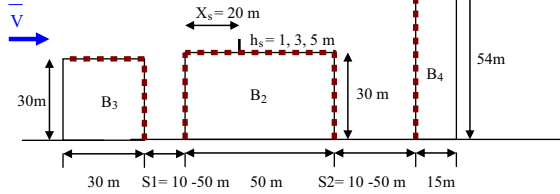


Fig. 7. Configurations 3 and 4: Building placed upstream and downstream of sources located on a low and intermediate height building.

6.1. Building placed upstream and downstream of a lower height building

The effects of a taller building placed upstream and downstream of a source located on a shorter building (Configuration 3) are presented in this section as shown in Figs. 8–10.

6.1.1. Dilutions on the leeward wall of the upstream building (B_3) for $X_s=0$

Fig. 8(a) shows normalised dilutions on the leeward wall of the taller upstream building for $X_s=0$ and $h_s=1$ m. Due to the recirculation zone created in the wake of the upstream building, as explained previously in Fig. 4, the plume is drawn towards the leeward wall of B_3 . The dilutions increase by a factor of about 4 for $M=1$ to $M=2$. This is because an increase in exhaust speed leads to greater dispersion of pollutants, a part of which remains engulfed within the wake of the upstream building, while a part of it escapes along the roof of the emitting building. At $h_s=3$ m the

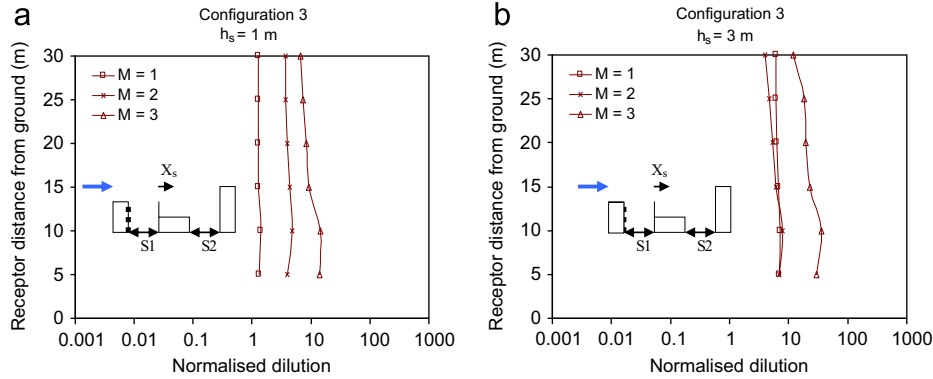


Fig. 8. Normalised dilution on leeward wall of B_3 for $X_s=0$ and $S_1=S_2=20$ m: (a) $h_s=1$ m; (b) $h_s=3$ m.

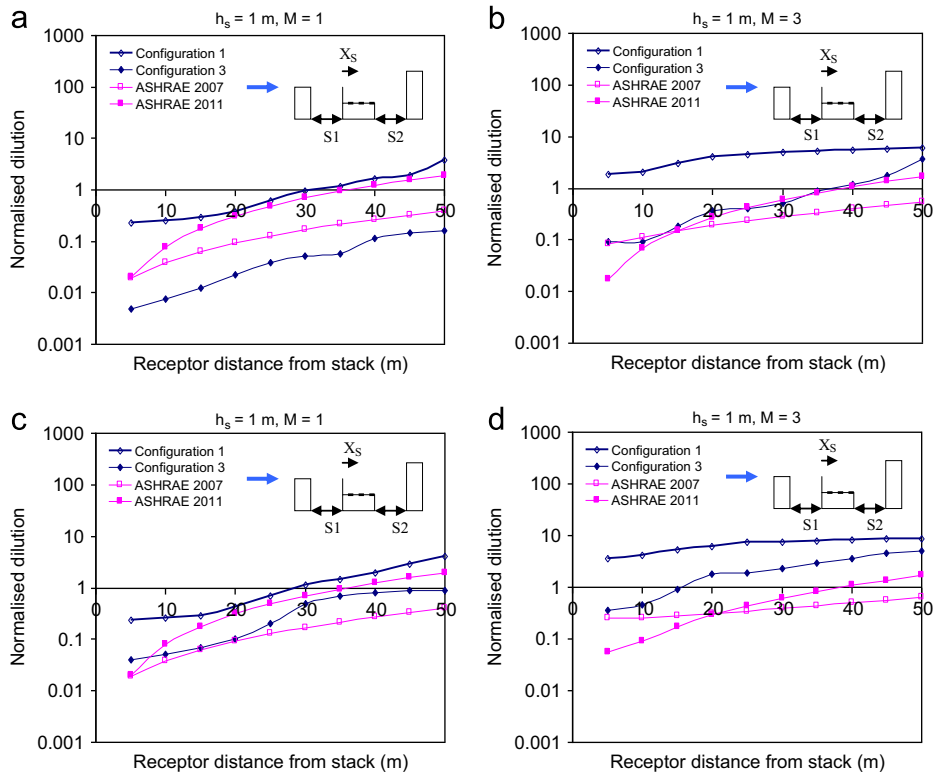


Fig. 9. Normalised dilution on rooftop of B_1 for $X_s=0$ and $S_1=S_2=20$ m: (a) $h_s=1$ m, $M=1$; (b) $h_s=1$ m, $M=3$; (c) $h_s=3$ m, $M=1$; (d) $h_s=3$ m, $M=3$.

plume tends to travel closer to the leeward wall of B_3 especially at $M=1$ and $M=2$ as shown in Fig. 8(b) leading to comparable dilutions at these M values. This is possibly because the plume rise increases due to greater stack height, coupled with the wake recirculation of B_3 . However, at $M=3$, an increase in exhaust speed leads to a further increase in plume height which makes the plume spread farther. This causes higher dilutions on the leeward wall compared to those obtained at $M=2$. In fact, at $h_s > 3$ m no concentrations were found on the wall possibly because the plume rise was high enough to overcome the wake recirculation of B_3 and affect only the emitting and downstream building. A similar trend was also observed for $X_s=20$ m although the dilutions were somewhat higher than their respective values at $X_s=0$. ASHRAE does not provide dilution estimates for the walls of a building and can be used only to assess roof dilutions on an emitting building. A close examination of the results obtained from a previous study of taller upstream building effects (see Hajra et al., 2011) reveals that the trends are nearly the same as in the present study. In other words, the presence of a taller downstream building (B_4) has a negligible effect on the dilutions on the leeward wall of the upstream building. However, this trend changes for roof dilutions on the emitting building, as discussed in the subsequent section.

6.1.2. Dilutions on rooftop of emitting building (B_1) for $X_s=0$

Fig. 9(a) shows dilutions on the rooftop of B_1 for $h_s=1$ m, $M=1$ and $X_s=0$ for Configurations 1, 3, ASHRAE, 2007, 2011. The dilutions predicted by Configuration 3 are almost 20 times lower than those predicted by Configuration 1 (isolated case). This is because the plume in the presence of an upstream building is brought closer to the roof of B_1 . Additionally, a portion of the plume gets trapped between the two buildings resulting in increased pollutant concentration. A similar trend is observed at $h_s=1$ m and $M=3$ as shown in Fig. 9(b) although the dilutions are somewhat higher than those produced at $M=1$ due to higher exhaust speeds. This trend remains unchanged for higher h_s and M as shown in Fig. 9(c) and (d). At $h_s=5$ m and $M=3$ the dilutions

obtained from Configurations 1 and 3 become comparable possibly because the plume rise is sufficient to overcome the effect of adjacent buildings. ASHRAE, 2007 predicts about 10 times lower dilutions than Configuration 1 due to low plume rise estimates, making the results overly conservative for $h_s=1$ m and $M=1$, while ASHRAE, 2011 predictions compare well with wind tunnel data for the isolated case, except at receptors close to the stack. This trend remains unchanged for $h_s=3$ m and $M=1$. However at higher M values ($M=3$), both the 2007 and 2011 versions predict lower dilutions than wind tunnel data for the isolated building. This is because unlike the 2007 version, which primarily estimates the plume spread parameters in terms of M , ASHRAE, 2011 estimates these values as a function of roughness length. The plume spread parameters in ASHRAE, 2011 do not change despite an increase in M . In general, both versions are incapable of assessing plume dilutions for adjacent building cases. A comparison of the present study with the results of a taller upstream building in the presence of a source located on a shorter building (see Hajra et al., 2011) reveals that the roof dilutions from the present study are much lower than upstream building configuration results. This is because in the present study, the tall building placed upstream and downstream of the low building inhibits the plume from dispersing freely, thereby increasing the gas concentrations.

6.1.3. Dilutions on the rooftop of emitting building (B_1) for $X_s=20$ m

Fig. 10(a) shows comparisons for Configurations 1, 3, ASHRAE, 2007, 2011 for $h_s=1$ m, $M=1$ and $X_s=20$ m. Configuration 3 predicts lower dilutions than Configuration 1 at all receptors. Additionally, Configuration 3 permits transport and dilution upwind of the stack due to the presence of an upstream building, as described previously in Fig. 4. The dilutions are particularly low closer to the stack because the plume strikes the leeward wall of the upstream building and travels back towards the roof of the emitting building increasing effluent concentrations closer to the centre of the roof. Thereafter, at receptors closer to the downwind edge of B_1 the dilutions become comparable to the isolated case

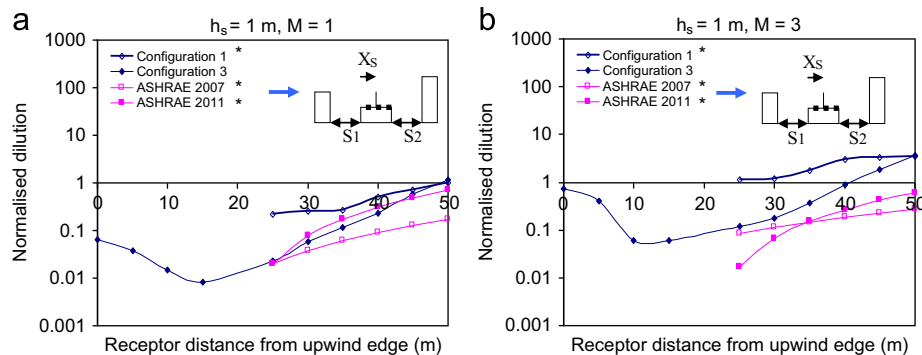


Fig. 10. Normalised dilution on rooftop of B_1 for $X_s=20$ m and $S_1=S_2=20$ m: (a) $M=1$; (b) $M=3$ (*Concentration of pollutant was found only downwind of the stack).

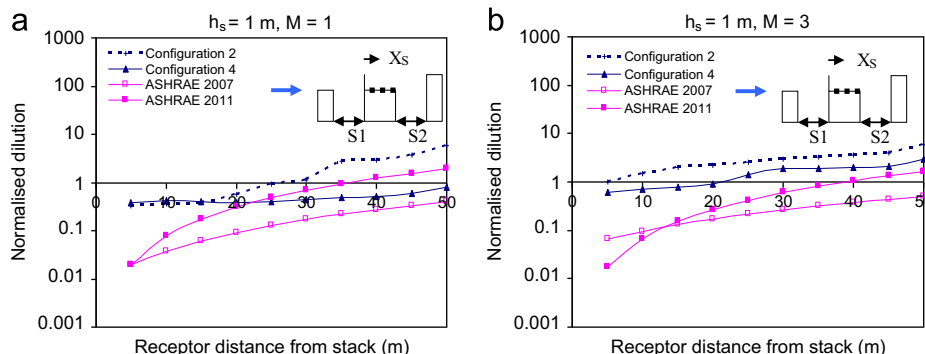


Fig. 11. Normalised dilution on rooftop of B_2 for $X_s=0$ and $S_1=S_2=20$ m: (a) $M=1$; (b) $M=3$.

because the effect of the upstream building reduces and some of the pollutants escape through side leakage. A similar trend is observed at $h_s = 1$ m and $M = 3$ as shown in Fig. 10(b) although the dilutions are somewhat higher than those at $M = 1$. At $h_s = 5$ m the dilutions obtained by Configurations 1 and 3 become comparable, although rooftop concentrations were obtained upwind of stack even for such cases. ASHRAE, 2007 predicts about 10 times lower dilutions than the isolated building (Configuration 1), while ASHRAE, 2011 estimates compare well at points beyond 30 m downwind of the stack for $h_s = 1$ m and $M = 1$. However, at $M = 3$ both the ASHRAE versions predict lower dilutions than the isolated building due to reasons explained previously. In general, ASHRAE, 2007, 2011 do not predict dilutions upwind of the stack because they do not take account of the turbulence caused by adjacent buildings. Taller upstream building configurations with centrally placed stacks also produce similar trends although the dilutions are somewhat higher than those obtained from the present study (Hajra et al., 2011).

6.2. Building placed upstream and downstream of an intermediate height building

The effects of a building placed upstream and downstream of a stack mounted on an intermediate height building (Configuration 4) are presented in Figs. 11 and 12.

6.2.1. Dilutions on rooftop of emitting building (B_2) for $X_s = 0$

Fig. 11(a) shows comparisons for Configurations 2 and 4 in terms of normalised dilutions on rooftop of the intermediate height building (B_2) for $h_s = 1$ m, $M = 1$ and $X_s = 0$. It was observed that Configurations 2 and 4 predict comparable dilutions at receptors located within 15 m downwind of stack. Thereafter, Configuration 4 predicts lower dilutions than the isolated case. This is because the upstream building is of equal height as the emitting building, which generates lower turbulence in the wake of the upstream building. Hence, the plume geometry assumes a shape similar to the isolated case (conical shape). However, the presence of the taller downstream building inhibits the plume from escaping thereby reducing roof dilutions at points beyond 15 m from the upwind edge. This trend changes at $h_s = 1$ m and $M = 3$ as shown in Fig. 11(b) where the dilutions predicted by Configuration 4 are lower than the isolated case by a factor of about 2. At $h_s > 1$ m comparable dilutions are obtained at all receptors for both configurations. Additionally, for centrally placed stacks the effect of adjacent buildings diminishes completely as dilutions at all downwind receptors are comparable to the isolated case. Also, unlike Configuration 3 discussed previously, there are no rooftop concentrations found upwind of stack due to equal heights of upstream and emitting buildings. It is not surprising to observe a similar trend for upstream configurations for buildings of similar height (see Hajra et al., 2011). This is because, at low stack heights, and due to the greater height of the emitting

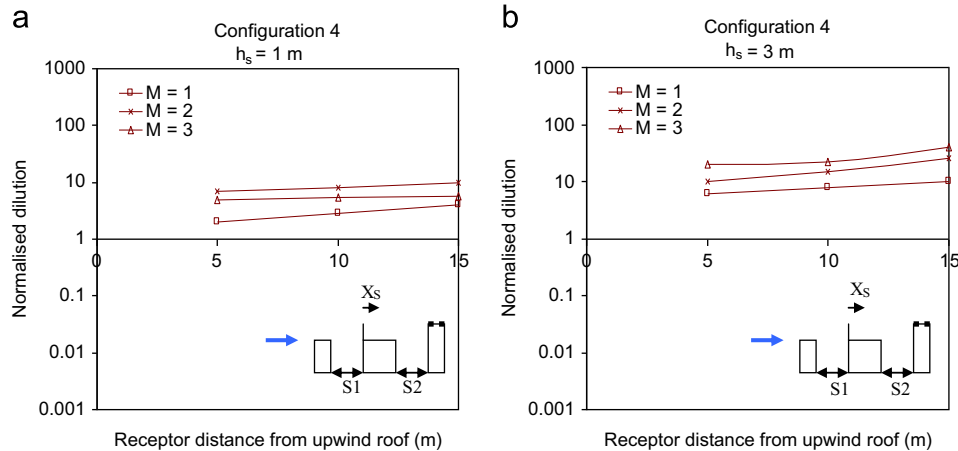


Fig. 12. Normalised dilution on rooftop of B4 for $X_s = 0$ and $S_1 = S_2 = 20$ m: (a) $h_s = 1$ m; (b) $h_s = 3$ m.

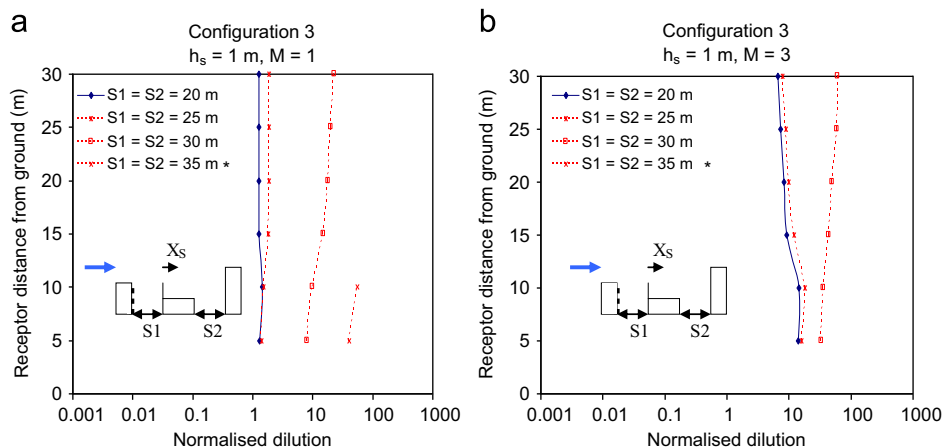


Fig. 13. Normalised dilution on leeward wall of B3 for different building distances and $X_s = 0$: (a) $M = 1$; (b) $M = 3$ (*Concentration of pollutant was detected only at two receptors at $M = 1$, while none of the receptors detected any concentrations at $M = 3$).

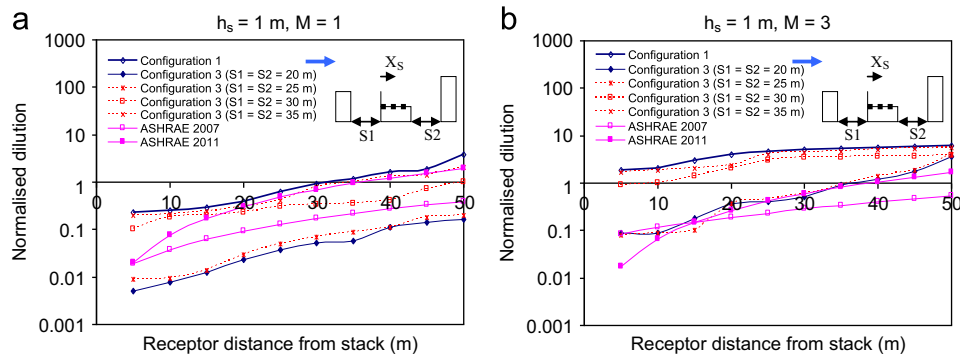


Fig. 14. Normalised dilution on rooftop of B1 for different building distances and $X_s=0$: (a) $M=1$; (b) $M=3$.

building, most of the pollutants would disperse freely as side-leakage (as shown in Fig. 5) and only a small part of the plume will remain closer to the roof due to the recirculation cavity of the upstream building. ASHRAE, 2007 and 2011 predict about 10 times and five times lower dilutions respectively than wind tunnel data for the isolated building (Configuration 1) for $h_s=1$ m and $M=1$, while at $h_s=1$ m and $M=3$ both ASHRAE versions predict lower dilutions than the isolated building.

6.2.2. Dilutions on the rooftop of downstream building (B_4) for $X_s=0$

Fig. 12(a) shows normalised dilutions on the roof of the downstream building (B_4) for Configuration 4 at $h_s=1$ m and $X_s=0$. It was observed that the dilutions were somewhat lower for $M=3$ than at $M=2$. This is because an increased exhaust speed deposits more pollutants on the roof of the downstream building due to upwash. Also, the height of the emitting building (B_2) is equal to the upstream building (B_3) thereby reducing any chances of the plume travelling towards the leeward wall of B_3 . This trend changes at $h_s=3$ m where the dilutions increase with an increase in M value due to greater plume spread with increased exhaust speeds (Fig. 12(b)). At $h_s > 3$ m no rooftop concentrations were found on the roof of B_4 since a major part of the plume affects the emitting building with a portion of it escaping through side leakage. Similar observations were also made for $X_s=20$ m although the dilutions were somewhat higher than those measured at $X_s=0$. These trends remain unchanged when compared to results from Hajra and Stathopoulos, 2012 for taller downstream building configurations. ASHRAE does not provide formulations to assess plume dilutions on the roof of the downstream building.

Sections 6.1 and 6.2 presented dilutions on various building surfaces for a spacing between buildings ($S_1=S_2$) equal to 20 m. The subsequent section describes the dilution results when the spacing between buildings exceeds 20 m.

6.3. Effect of spacing between the buildings

The effect of spacing between buildings is discussed in Figs. 13–15 for Configuration 3, as this particular case was found to be more critical than Configuration 4. As the spacing between buildings exceeded 20 m the roof dilutions on the intermediate building (B_2) became comparable to the isolated case (Configuration 2). This is because the upstream building (B_3) is of equal height to the emitting building (B_2) in Configuration 4, and hence the effect of spacing on the dilutions was negligible. It may be noted that the spacing between upstream and emitting building (S_1), and downstream and emitting building (S_2) were varied uniformly.

6.3.1. Dilutions on leeward wall of upstream building (B_3)

Fig. 13(a) shows the effect of spacing between the buildings for Configuration 3 on the leeward wall of B_3 at $h_s=1$ m, $M=1$ and $X_s=0$. It was observed that at a spacing of 20 m and 25 m comparable dilutions were obtained on the leeward wall of the upstream building. This is because due to the recirculation length in the wake of the upstream building a large portion of the plume remains trapped and despite a change in spacing (increase in 5 m) the dilutions remain unchanged. However, this trend changes at a spacing of 30 m where the dilutions are almost 10 times higher than those obtained at a spacing of 20 m. This is because the buildings are sufficiently away from the wake recirculation of the upstream building resulting in less plume material being engulfed. At a spacing of 35 m the dilutions are even greater resulting in plume concentrations only on the first two receptors closer to the ground. A similar trend is observed at $h_s=1$ m and $M=3$ as shown in Fig. 13(b) where although dilutions at a spacing of 20 m and 25 m are comparable, they are higher at a spacing of 30 m. In fact, at spacing greater than 30 m, no plume concentrations were found on the leeward wall as most of the plume affects only the emitting and downstream buildings.

6.3.2. Dilutions on rooftop of emitting building (B_1)

Comparable rooftop dilutions on emitting building (B_1) at a spacing of 20 m and 25 m were found at $h_s=1$ m, $M=1$ and $X_s=0$ for Configuration 3 although the dilutions were about 10 times lower than the isolated case as shown in Fig. 14(a). At a spacing of 30 m the dilutions increase by a factor of about 10, although at points closer to the leeward edge the dilutions were lower than the isolated case. This is because at distances between 20 m and 25 m the plume geometry does not change markedly. At spacing greater than 30 m the dilutions gradually become closer to the isolated case because the shape of the plume tends to be more conical, similar to that of an isolated building. Similar trends were observed for $h_s=1$ m, $M=3$ as shown in Fig. 14(b), although the dilutions were somewhat higher than those obtained at $M=1$. ASHRAE, 2007, 2011 are not capable of incorporating the effect of spacing between buildings and generate dilutions only for an isolated case, as discussed previously.

6.3.3. Dilutions on windward wall of downstream building (B_4)

Fig. 15(a) shows the effect of spacing between the buildings for Configuration 3 on the windward wall of B_4 at $h_s=1$ m, $M=1$ and $X_s=0$. It was observed that when the buildings were spaced 20 m and 25 m apart, comparable dilutions were obtained on the windward wall of the downstream building due to reasons explained previously. At a spacing of 30 m dilutions were found only at receptors close to the ground as most of the plume escapes

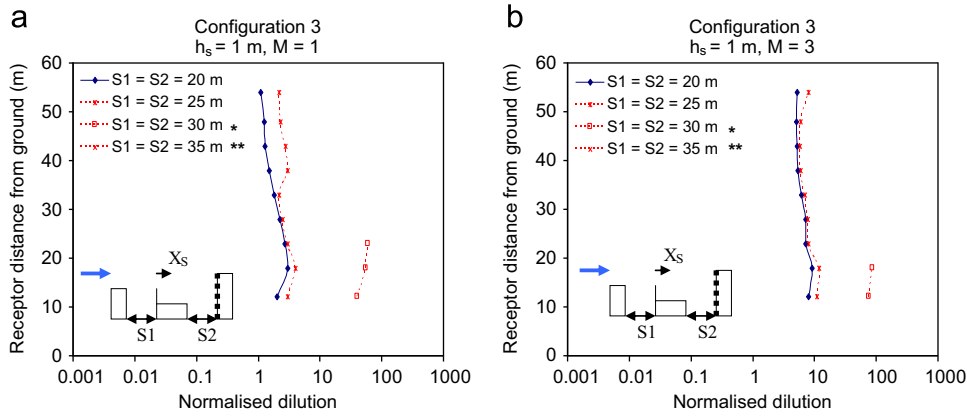


Fig. 15. Normalised dilution on windward wall of B4 for different building distances and $X_s=0$: (a) $M=1$; (b) $M=3$ (*Concentration of pollutant was found only at few receptors; **Concentrations were undetectable at all receptors).

through side-leakage. A similar trend is observed at $h_s=1$ m and $M=3$ as shown in Fig. 15(b) where although dilutions at a spacing of 20 m and 25 m are comparable, they are higher than those obtained at $M=1$ due to greater exhaust speeds. In general, as the spacing between the buildings exceeded 30 m, no plume concentrations were found on the windward wall of B₄.

7. Design guidelines and summary of the present research

This section presents design guidelines for safe placement of stack and intake on various building surfaces, as well as a summary of the research findings.

7.1. Design guidelines

Based on near-field pollutant dispersion experiments, design guidelines for the safe placement of stack and intake on various building surfaces have been summarised as follows.

7.1.1. Building placed upstream and downstream of a low height building

1. When the emitting building lies within the recirculation length of the upstream building (L_{r1}), and the downstream building lies within the recirculation length of the emitting building (L_r), intakes should not be placed anywhere on the emitting building, leeward wall of the upstream building and windward wall of the downstream building, irrespective of stack height and location (Fig. 16(a)).
2. When the buildings are spaced sufficiently apart such that each building is beyond the wake recirculation of the upstream building (Fig. 16(b)), intakes may be placed on all building surfaces for any stack location. However, for low stacks with low exit velocities, intakes should be avoided very close to the stack and leeward wall of the emitting building.

7.1.2. Building placed upstream and downstream of an intermediate height building

1. When the buildings are closely spaced wherein the influence of the wake recirculation of any given building will affect the building placed downstream of it, irrespective of stack location and height, intakes may be placed anywhere on the upstream building surface. However, intakes should be avoided close to the upwind edge of the downstream building.

2. When the buildings are spaced sufficiently apart (as in Fig. 16 (b)), the provisions for a source placed on a low building for placement of intakes (Fig. 16(b)) may be used.

7.2. Summary of the results of the present study

The results of the experimental study can be summarised as follows:

7.2.1. Effect of the height of the building where the stack is located

1. When a tall building is placed upstream and downstream of a lower building, then a part of the plume gets engulfed within the recirculation zone of the upstream building, and a part of it remains trapped between the two buildings, thereby increasing rooftop concentrations.
2. A low building in the presence of a taller upstream and a taller downstream building also affects the leeward wall of the upstream building and the windward wall of the downstream building for stack heights less than about 5 m.
3. When the height of the building where the stack is located is equal to the upstream building, a part of the plume accumulates on the roof of the downstream building, besides affecting the emitting building for low stack heights.
4. A stack located on a building in the presence of an upstream building of equal height and a taller downstream building generates comparable rooftop dilutions on that building to that of the isolated case, for stack heights greater than about 3 m irrespective of M .

7.2.2. Effect of spacing ($S1$ or $S2$) between the buildings

1. As the spacing between the shorter building with a rooftop source and the taller upstream building, exceeds the recirculation zone of the latter, dilutions on the roof of the emitting building gradually become comparable to the isolated case for any h_s and M value.
2. Similarly, as the spacing between the taller downstream building exceeds the recirculation zone of the shorter building with a rooftop pollutant source, the windward wall of the downstream building is less affected irrespective of stack height.
3. When the height of the emitting building equals the height of the upstream building, spacing does not influence the dilutions on the various building surfaces due to the height of the

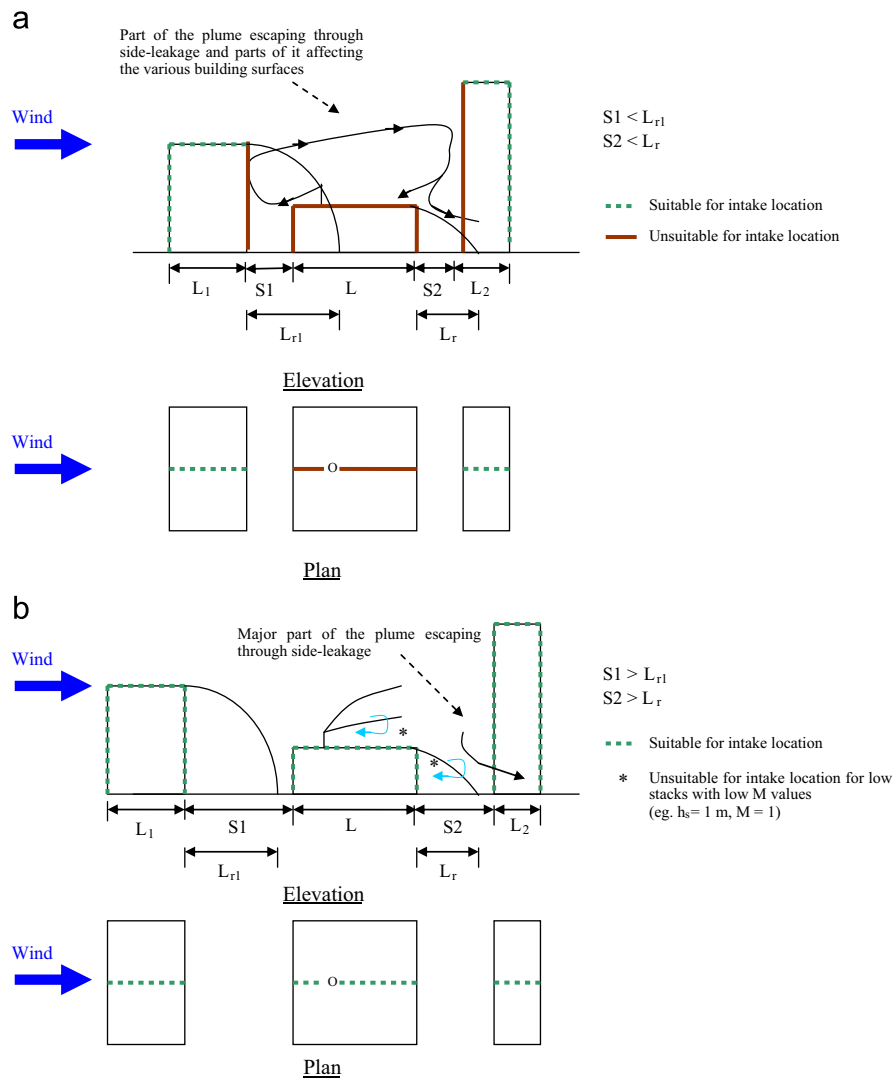


Fig. 16. Schematic representation for suitability of intake location at various building surfaces: (a) $S1 < L_{r1}$ and $S2 < L_r$; (b) $S1 > L_{r1}$ and $S2 > L_r$.

emitting building which allows the plume to spread further thereby making the dilutions comparable to the isolated case.

formulations be re-visited; in particular the effects of spacing between buildings and local topography must be incorporated.

7.2.3. ASHRAE

1. ASHRAE, 2007 generally predicts lower dilutions than experimental data for all configurations, while ASHRAE, 2011 predictions compare well with wind tunnel data for the isolated building only for low M values ($M=1$).
2. ASHRAE, 2011 restricts the value of ζ to zero when $h < H_c$, which reduces the value of the exponential term in Eq. (8), resulting in lower dilutions. The dilutions predicted by ASHRAE, 2011 could possibly be improved by modifying the conditions that result in the value of ζ , thereby reducing the discrepancy between ASHRAE, 2011 and experiment.
3. In general, ASHRAE considers only the across wind dimension and the height of the emitting building to estimate the dilutions on the rooftop of the emitting building, without considering the effect of turbulence caused by the building and its surroundings.
3. ASHRAE is incapable of assessing plume dilutions on the wall of a building and can only estimate dilutions on rooftop of isolated buildings. Therefore, it is suggested that ASHRAE

8. Conclusions

The present study investigated the near-field plume dispersion characteristics of two multiple-building configurations in the Boundary Layer Wind Tunnel Laboratory at Concordia University. Results show that when a tall building is placed upstream and downstream of a low building with a rooftop stack, the plume affects the leeward wall of the upstream building, roof of the emitting building and the windward wall of the downstream building. However, when the upstream building is of similar height to the emitting building, only the windward wall of the downstream building and the roof of the source building are affected negatively. In general, the spacing between buildings and the heights of each building are critical factors affecting the plume dilutions. ASHRAE, 2007 predicts lower dilutions than wind tunnel data for the isolated building, while ASHRAE, 2011 predictions compare well with wind tunnel data for a few cases. In the future, the ASHRAE model should review the conditions that lead to the value of ζ , in order to obtain reasonable dilution estimates. Guidelines for the safe placement of stack and intake on various

Table 3
Summary of calculations following ASHRAE 2007, 2011 for Fig. 9(a).

Parameter	Value used	Remark
h_s	1 m	Chosen value of stack height pertaining to Fig. 9(a)
d_e	0.6 m	Stack diameter
M	1	Exhaust momentum (V_e/U_H)
U_H	6.2 m/s	Wind speed at building height H , where $H=15$ m
h_d	1.2 m	Plume downwash from Eq. (7)
V_e	6.2 m/s	Exhaust velocity
t_{avg}	15 min	Recommended by ASHRAE, 2011
β	1	Value for an uncapped stack
L_r	22.31 m	Building recirculation length from Eq. (1)
H_e (or h_{top})	4.91 m	Height of recirculation zone from Eq. (2)
Q	1.753 m ³ /s	Discharge rate of effluents from stack ($\pi \times 0.25 \times 0.6^2 \times V_e$)

building surfaces to prevent pollutant re-entry have been suggested based on this study.

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Appendix

For the low-rise building considered in this study (refer to Fig. 9 (a)), the receptor lying 20 m downwind of the stack has been chosen. Table 3 presents a summary of the calculations, which are common to both ASHRAE versions.

ASHRAE, 2007

ASHRAE, 2007 defines a term called “ ζ ”, which is the vertical separation between plume height (h) and h_{top}

Plume rise (h_r) = 1.8 m (from Eq. (6))

$h = h_s + h_r - h_d = 1.6 \text{ m} < h_{top}$

$\therefore \zeta = 0$

At $X=20$ m

$\sigma_y/d_e = 4.675$ (from Eq. (9));

$\sigma_z/d_e = 3.500$ (from Eq. (10));

$D_r = 65.465$ (from Eq. (8));

$D_{normalised} = 0.0916$ (from Eq. (11)) – see value in Fig. 9(a).

ASHRAE, 2011

The plume rise is found from a series of calculations as described further:

Assume $Z_o = 2$ m for an urban terrain (from Table 1, ASHRAE, 2011, Chapter 45)

$U_H/U^* = 5.03$ (from Eq. (16));

$h_f = 0.455$ m (from Eq. (14));

$h_x = 1.451$ m (from Eq. (13));

$\therefore h_r = 0.455$ m (from Eq. (12));

$h = 0.255$ (from Eq. (5))

Since, $h < h_{top}$

$\therefore \zeta = 0$

ASHRAE, 2011

$i_x = 0.363$ (from Eq. (21));

$i_y = 0.273$ (from Eq. (19));

$i_z = 0.182$ (from Eq. (20));

$\sigma_y = 5.464$ (from Eq. (17));

$\sigma_z = 3.646$ (from Eq. (18));

$D_r = 221.35$ (from Eq. (8));

$D_{normalised} = 0.278$ (from Eq. (11)) – see value in Fig. 9(a).

ASHRAE, 2011 also states that the calculations should be repeated for $0.5Z_o$ and $1.5Z_o$, and the lowest dilution must be considered for the design. For the present study, an urban terrain was considered ($Z_o = 2$ m), and it was found that dilutions at $0.5Z_o$ and $1.5Z_o$ would have made negligible changes. Therefore, ASHRAE, 2011 dilution results were found for $Z_o = 2$ m.

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